Integral-root carriers for Con(ZFC)

A degree-24 equation in fifteen positive unknowns, and a degree-629 companion in twelve

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Abstract

We give a fixed integer polynomial in fifteen strictly positive unknowns, of exact total degree 24, whose lack of positive zeros is provably equivalent in ZFC to syntactic Con(ZFC). Its division-free evaluator uses 90 live binary arithmetic operations. The comparable degree frontier in the supplied ProveIt material is degree 44 at 75 operations. Two further tradeoffs have degree/operation pairs $(26, 88)$ and $(30, 79)$. A twelve-positive-variable companion has degree 629 at 114 operations, improving the supplied degree-897, 108-operation point.

The new reduction replaces expensive coordinate scales by divisibility forced through existing monic square equations. A rational offset is placed inside a square root and denominators are cleared; integrality then recovers the missing scale without an extra unknown. Positive Pell gaps supply the reverse construction. Reducing the auxiliary expression modulo an explicit index equation lowers degree 26 to 24.

The fixed coefficient tuple, calibrated compiler, and standard Pell lemmas are inherited with precise provenance. The new reductions are proved here by ordinary mathematics formalizable in ZFC; they are not a newly checked end-to-end proof-assistant development. Literal circuits, exact algebra certificates, independent finite checks, coefficient payloads, and source proofs accompany the paper. No global minimum, consistency proof, or new operation-count record is asserted.

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1 Result, comparison, and proof boundary

The goal is a particularly simple *fixed* Diophantine representation of ZFC inconsistency. The polynomial is not a family with an unspecified machine parameter: the four numeral ports used below are bound to explicit finite binary integers in the archive. Every other abbreviation is a polynomial expression in the listed unknowns.

Theorem 1.1 (Principal result)**.** *With the coefficient tuple specified in Section 10, the polynomial $P_{15,24}$ of Section 3 has fifteen positive unknowns, exact total degree* 24*, and a* 90*-operation evaluator. At the ordinary mathematical proof level of the pinned native compiler,*

$$\mathrm{ZFC} \vdash \left[\neg\exists \mathbf{x} \in \mathbb{Z}_{>0}^{15}\ P_{15,24}(\mathbf{x}) = 0\right] \longleftrightarrow \mathrm{Con(ZFC)}.$$

The polynomials $P_{15,26}$ and $P_{15,30}$ have the same equivalence, with degree/operation pairs $(26, 88)$ *and* $(30, 79)$*. The companion P_{12} has twelve positive unknowns, degree* 629*, and* 114 *operations.*

Here Con(ZFC) is the syntactic assertion that the fixed effective ZFC proof calculus has no finite refutation. The inherited K215 calibration uses that assertion, and its ordinary internalization supplies the displayed provability claim. This notation does not claim that a new object-language ZFC proof file or a proof-assistant certificate has been produced.

There are several distinct notions of simplicity. Table 1 keeps the domain, number of supplied unknowns, total degree, and literal operation count separate. In particular, a reduction of degree at a fixed number of positive unknowns should not be described as a reduction of the number of unrestricted integer unknowns.

Construction	Positive unknowns	Degree	Operations	Status
Source 47, minimum-norm grouping	15	44	75	Prior
Source 49, two root gaps	15	44	76	Prior
Source 51, sparse product	15	157	66	Prior
$P_{15,24}$	15	**24**	90	New
$P_{15,26}$	15	**26**	88	New
$P_{15,30}$	15	**30**	79	New
Source 52, lean norm	12	897	108	Prior
P_{12}	12	**629**	114	New

Table 1: Bounded comparisons with the supplied source collection. All entries are single equations; systems are not counted as single polynomials.

Degree 24 is 45.45% below 44, with fifteen more arithmetic gates; degree 30 is 31.82% below 44, with four more gates. The twelve-variable degree reduction is 29.88%, with six more gates. These points make a substantial improvement on one coordinate of the measured frontier. They do not dominate the 66-operation construction or the constructions with fewer variables.

The read-only comparison includes the incoming wave-7 archives and their subsequent source directories 46–53. The working snapshot is ProveIt commit `6d375f2d2deac0519e659bbce5b48087023c1430`; retrieval started at `4b4349064a750ba6347293b55874a7f92abdd0db`. The archive records individual source hashes. Source 47 supplies the 44/75 grouping, and source 49 supplies the directly related 44/76 root-gap point, source 51 supplies the squared-modulus quotient used below, and source 52 supplies the twelve-variable predecessor [3, 2, 4, 5]. The broader small-variable context includes Sun's ten-integer-unknown preprint and the work on eleven integer unknowns and Isabelle formalization [7, 8, 9]. Neither result is needed for our fifteen-positive-variable proof.

The semantic dependencies are explicit. First, a finite refutation is equivalent to a blank K215 halting computation. Second, a native compiler encodes that computation with the authentic four-port tuple. Third, the ordinary Pell kernel encodes the resulting arithmetic constraints. The

new work is an existence-equivalence reduction from that established kernel to the equations here. According to the preserved K215 status document, the exact machine endpoint already has compiled Lean and Rocq correctness proofs [6]. We did not rerun those builds. The remaining formalization boundary for this contribution is the native Diophantine compiler together with the new full equation and its internalization.

2 The mechanism: force a scale through an integral root

Lemma 2.1 (A rational square root of an integer is integral)**.** *If $m > 0$ and $m^2 \mid n^2$ for integers m, n, then $m \mid n$. Equivalently, a rational number whose square is an integer is an integer.*

Proof. For every prime p, $2v_p(n) \geq 2v_p(m)$ implies $v_p(n) \geq v_p(m)$. The case $n = 0$ is immediate. Alternatively reduce the rational fraction to lowest terms and compare denominator prime factors. □

Suppose a construction uses a scaled coordinate $Y = sq^3$ and a square equation $\tau^2 = A$. If the canonical witnesses have $\tau > L + s$, one can replace the supplied coordinate s by Y and the supplied root τ by a positive gap r. The polynomial

$$[q^3(L + r) + Y]^2 = q^6 A$$

forces $q^3 \mid Y$ by Lemma 2.1, and then restores $\tau = L+r+Y/q^3$. The converse uses $r = \tau-L-s > 0$.

The important issue is not the elementary divisibility lemma by itself. One must find an existing root that can carry the fractional offset, prove that no extra roots are introduced, and prove a sufficiently large positive margin for the converse. Otherwise a change of variables can silently exclude all intended witnesses. Here the first Pell norm supplies one such margin, and the main Pell norm supplies another.

The second application is sequential. After recovering $q^3 \mid Y$, we know $H \equiv 3 \pmod{q}$. A canonical congruence restriction gives $\gcd(q, 3) = 1$. Therefore

$$(qD_0 + XH)^2 = q^2(1 + \Delta c^2) \quad \Longrightarrow \quad q \mid XH \quad \Longrightarrow \quad q \mid X.$$

Thus both formerly scaled coordinates can be supplied directly. The divisibility conditions have been transferred into equations already needed for the construction, without introducing quotient unknowns.

The final degree reduction uses a separate idea: after imposing $R = k - hXY - 1$, use the lower-degree right side inside the auxiliary norm. This is an exact reduction modulo a retained equation, not a heuristic approximation to R.

3 The explicit degree-24 equation

The fifteen unknowns, all in $\mathbb{Z}_{>0}$, are

$$j, F, \alpha, z, f, h, i, T, X, Y, r, \eta, \zeta, u, \gamma.$$

The four fixed integers are β, K_0, M_C, M_F. In particular $\beta = 2^{590069} - 1$, and M_F denotes the already shifted source port. Define

$$\begin{aligned}
J &= 3j + 1, & q &= \beta J + 1, & C &= q - F - \alpha,\\
E &= XY, & k &= \eta + \zeta, & c &= kY + \eta,\\
a &= E + Y, & H &= 4a + 3, & \Delta &= a^2 + H = (a+2)^2 - 1,\\
D_0 &= X + ac + \gamma H, & \ell &= EkY, & I &= ic^2,\\
G &= (q-1)(q-F) + \alpha,
\end{aligned}$$

$$\begin{aligned}
R &= G(q^2-1) + (M_C + qM_F)J,\\
\widehat{R} &= k - hE - 1,\\
U &= \Delta(f^2 - 1),\\
V &= f^2(cT - \widehat{R}) - c,\\
W &= q\big[K_0 C + (q-F) - z(q-1)\big] + XC.
\end{aligned}$$

Set the six residuals

$$A_0 = [q^3(\ell + r) + Y]^2 - q^6[\ell(\ell + k) + 1], \tag{1}$$
$$A_1 = (qD_0 + XH)^2 - q^2(\Delta c^2 + 1), \tag{2}$$
$$A_k = \widehat{R} - R, \tag{3}$$
$$A_t = W - q, \tag{4}$$
$$A_a = (V + u)^2 - Uu(2V + u) - 1, \tag{5}$$
$$A_s = I^2 - U. \tag{6}$$

The requested fixed polynomial is

$$\boxed{P_{15,24} = A_0^2 + A_1^2 + A_k^2 + A_t^2 + A_a^2 + A_s^2 = 0.} \tag{7}$$

This is an ordinary polynomial equation over the integers. There are no variable exponents, divisions, radicals, bit operations, or implicit existentially quantified registers in its definition. The named expressions only specify a compact division-free arithmetic circuit for that polynomial. The largest exponent in the notation is a fixed numeral.

The extra roots that could arise from allowing negative supplied unknowns are not covered by this theorem. Section 11 gives explicit translations when the intended domain is nonnegative or unrestricted integers.

4 The inherited kernel used by the proof

We state the precise interface rather than assume an unspecified "universal Diophantine polynomial." Let the old positive coordinates be

$$J, F, \alpha, z, f, h, i, T_{\mathrm{old}}, s, w, \tau, \eta, \zeta, y, \gamma_{\mathrm{old}}.$$

Use the same abbreviations as above, except

$$X = wq, \quad Y = sq^3, \quad D = X + ac + \gamma_{\mathrm{old}} H, \quad V_{\mathrm{old}} = c(T_{\mathrm{old}} f - 1) - Rf^2.$$

The ordinary source factors are

$$\begin{aligned} N_0 &= \tau^2 - \ell(\ell + k), & N_1 &= D^2 - \Delta c^2, \\ N_a &= U(V_{\mathrm{old}}^2 - y^2) + y^2, & N_k &= k - hE - R, \\ N_t &= (K_0 + w)C + (q - F) - z(q - 1), & N_s &= 1 + i^2 c^4 - U. \end{aligned}$$

The inherited ordinary proof establishes soundness of

$$N_0 N_1 N_a N_k N_t N_s = 1 \tag{8}$$

on these positive coordinates: a solution yields a finite ZFC refutation. The proof handles the possible unit signs before recovering $N_k = N_t = 1$; it does not simply assume each factor equals $+1$. Its complete ordinary argument is preserved as `upstream/ORDINARY_KERNEL.md`.

Conversely, a finite refutation gives canonical normalized witnesses with all these factors equal to $+1$. The ordinary strong coordinate is obtained from a normalized coordinate i_0 by

$$i = \Delta i_0, \qquad f^2 = 1 + \Delta i_0^2 c^4, \qquad I = ic^2 = \Delta i_0 c^2, \qquad U = I^2. \tag{9}$$

The authentic compact adapter also supplies

$$q = B^N, \qquad J = 1 + B + \cdots + B^{N-1}, \qquad B = 2^{590069}, \quad N = 581197^n > 1, \tag{10}$$

$$R \equiv 3 \pmod 4, \qquad c = \psi_{a+2}(R), \qquad \tau = \chi_{P_0}(n_0), \quad k = 2\psi_{P_0}(n_0), \tag{11}$$

$$P_0 = 2XY^2 + 1, \qquad n_0 = (R+1)/2 \geq 3, \qquad k > XY, \quad q \geq 16.$$

The exponent n in (10) and the Pell index n_0 are proof-only integers, not supplied unknowns. Padding lets the canonical exponent be taken arbitrarily large as required by the compiler.

Here the Pell polynomials satisfy

$$\chi_A(0) = 1, \quad \psi_A(0) = 0, \qquad \chi_A(n) + \psi_A(n)\sqrt{A^2 - 1} = (A + \sqrt{A^2 - 1})^n.$$

They have integer coefficients in A. Their elementary recurrences, classification, strong divisibility, and index-recovery lemmas are preserved in `upstream/PELL_LEMMAS.md` and the ordinary kernel proof.

The coefficient interface matters. Source 53's relaxed-rank exposition and source 51's quotient proof were originally distributed with different historical tuples. We use their coefficient-independent algebraic lemmas, and use source 44's *whole* compact tuple and its complete native adapter. In particular, odd N and $R \equiv 3 \pmod 4$ come from the adapter actually bound here, not from a borrowed numeric instance.

5 Soundness of the degree-24 construction

Suppose a positive tuple satisfies (7). Since squares of integers are nonnegative, all six residuals vanish.

5.1 Recover the two coordinate scales

Equation (1) and Lemma 2.1 give

$$q^3 \mid q^3(\ell + r) + Y, \qquad q^3 \mid Y.$$

Thus

$$s = Y/q^3 \in \mathbb{Z}_{>0}, \qquad \tau = \ell + r + s \in \mathbb{Z}_{>0}, \qquad \tau^2 - \ell(\ell + k) = 1. \tag{12}$$

The odd exponent 590069 gives $B \equiv 2 \pmod 3$ and $\beta \equiv 1 \pmod 3$. The supplied parameterization $J = 3j + 1$ consequently gives $q \equiv 2 \pmod 3$, without assuming any decoded form for q. Moreover $q \mid Y$ gives $H = 4Y(X + 1) + 3 \equiv 3 \pmod q$. Hence

$$\gcd(q, H) = \gcd(q, 3) = 1.$$

Equation (2) and the same lemma yield $q \mid XH$, and therefore

$$w = X/q \in \mathbb{Z}_{>0}, \qquad \gamma_{\text{old}} = \gamma + w \in \mathbb{Z}_{>0}, \qquad D = D_0 + wH. \tag{13}$$

After division by q^2, (2) is exactly $D^2 - \Delta c^2 = 1$. No prime, power-of-two, or squarefree hypothesis about an arbitrary supplied q has been used.

5.2 Recover the index, transport, and auxiliary witnesses

The index equation gives $R = \widehat{R}$, hence $N_k = 1$. Using $X = qw$, the transport numerator factors exactly as

$$W = q\big[(K_0 + w)C + (q - F) - z(q - 1)\big] = qN_t.$$

Since $W = q$ and $q > 0$, $N_t = 1$.

Put $T_{\text{old}} = fT > 0$. Then

$$V_{\text{old}} = c(T_{\text{old}} f - 1) - Rf^2 = f^2(cT - R) - c = V.$$

Let $y = |V + u|$. The exact identity

$$U\big[V^2 - (V + u)^2\big] + (V + u)^2 = (V + u)^2 - Uu(2V + u) \tag{14}$$

shows $N_a = 1$. We must also check $y > 0$. From $A_s = 0$, $U = I^2$ with $I = ic^2 > 1$; indeed $k \geq 2$ and $c = kY + \eta \geq 3$. If $y = 0$, the auxiliary norm would give $I^2V^2 = 1$, impossible for integer V and $I > 1$. Thus the restored ordinate is positive. The strong equation also gives $f > 1$ and $N_s = 1$.

We have reconstructed a positive solution of (8), in fact with all six source factors $+1$. The inherited soundness theorem therefore supplies a finite ZFC refutation. All divisions above occur only after their integrality has been proved; none occurs in the polynomial.

6 Completeness: the positive gaps really exist

Start with a finite ZFC refutation and choose the canonical normalized witnesses from Section 4. Retain the outer coordinates and set $i = \Delta i_0$ as in (9). We construct positive j, r, γ, T, u explicitly.

6.1 The cofinal congruence restriction

Both 590069 and $N = 581197^n$ are odd. Reducing the geometric sum in (10) modulo 3 gives

$$J \equiv 1 - 1 + 1 - \cdots + 1 \equiv 1 \pmod 3.$$

As $N > 1$, $J > 1$. Therefore $j = (J - 1)/3$ is a positive integer. This proves that imposing $J = 3j + 1$ retains the canonical witnesses; it does not claim this congruence for every old, noncanonical solution.

6.2 The first root has room for its offset

The old first norm is $\tau^2 = \ell(\ell + k) + 1$. Since $\ell = XY^2k > k$ and $k \geq 2$, it gives $\tau < \ell + k$. Consequently

$$\tau - \ell = \frac{\ell k + 1}{\tau + \ell} > \frac{k}{3}. \tag{15}$$

Canonical $k > XY$, $X \geq q \geq 16$ now imply

$$\frac{k}{3} > \frac{XY}{3} > \frac{Y}{q^3} = s.$$

Thus $r = \tau - \ell - s$ is a positive integer, and (1) holds with the new directly supplied $Y = sq^3$.

6.3 The main root has room for its offset

The old main root satisfies $D = \sqrt{\Delta c^2 + 1} > (a + 1)c$. Therefore

$$\gamma_{\text{old}} = \frac{D - ac - X}{H} > \frac{c - X}{H}. \tag{16}$$

With $P_0 = 2XY^2 + 1$ and $n_0 \geq 3$, the canonical first Pell coefficient satisfies

$$k = 2\psi_{P_0}(n_0) \geq 2\psi_{P_0}(3) = 8P_0^2 - 2 > 32X^2Y^4.$$

Hence $c > kY > 32X^2Y^5$. As $X \geq 16$ and $Y \geq 1$, we have $H = 4XY + 4Y + 3 \leq 9XY$. These bounds give $c - X > XH$, so

$$\gamma_{\text{old}} > X \geq X/q = w.$$

The new coordinate $\gamma = \gamma_{\text{old}} - w$ is therefore positive. Substituting it gives $D_0 = D - wH$, and hence (2). This strict margin is essential: merely observing that γ_{old} is positive would not justify subtracting w.

6.4 A squared-modulus quotient and the auxiliary gap

The squared-modulus quotient is an established ingredient from source 51, reproved here to make the completeness step explicit [4]. For $r_0 \geq 0$ there is an integer polynomial Q_{r_0} such that

$$\chi_t(2r_0 + 1) = tQ_{r_0}(t^2), \tag{17}$$

$$Q_{r_0}(0) = (-1)^{r_0}(2r_0 + 1), \qquad Q_{r_0}(1 - A^2) = (-1)^{r_0}\psi_A(2r_0 + 1). \tag{18}$$

Indeed $Q_0(x) = 1$, $Q_1(x) = 4x - 3$, and

$$Q_{r_0+1}(x) = (4x - 2)Q_{r_0}(x) - Q_{r_0-1}(x).$$

The first identity follows from the odd subsequence of the Pell recurrence. The evaluation at 0 follows by induction. Substitution $x = 1 - A^2$ negates the coefficient $4A^2 - 2$ of the odd ψ_A subsequence; its initial values are $1, 4A^2 - 1$, proving the second identity.

Write $A = a + 2$ and $R = 2r_0 + 1$. Canonical $R \equiv 3 \pmod 4$ makes r_0 odd. Define

$$V_* = \chi_I(R)/I = Q_{r_0}(I^2), \qquad y_* = \psi_I(R). \tag{19}$$

These are positive integers, and the Pell identity gives

$$I^2V_*^2 - (I^2 - 1)y_*^2 = 1. \tag{20}$$

Since $I^2 = \Delta(f^2 - 1) \equiv 1 - A^2 \pmod{f^2}$ and $c \mid I$, (18) gives

$$V_* \equiv -c \pmod{f^2}, \qquad V_* \equiv -R \pmod{c}.$$

Moreover (9) gives $f^2 \equiv 1 \pmod{c}$, so $\gcd(c, f^2) = 1$. It follows that

$$T = \frac{V_* + c + Rf^2}{cf^2} \in \mathbb{Z}_{>0}. \tag{21}$$

For this T, the new formula for V equals V_* once the index residual vanishes. This is a fresh auxiliary witness construction; it does not claim that every arbitrary old quotient was divisible by f.

Finally $R \geq 3$ and $I > 1$ imply $V_* > 1$. Rearranging (20) yields

$$y_*^2 - V_*^2 = \frac{V_*^2 - 1}{I^2 - 1} > 0.$$

Thus $u = y_* - V_*$ is a positive integer. With $V + u = y_*$, (14) proves $A_a = 0$. All the other residuals are zero by construction, including the unchanged canonical index and transport conditions. This completes both existence implications for $P_{15,24}$.

7 Exact degree and literal arithmetic cost

All fifteen supplied unknowns have degree one; the four fixed ports have degree zero. Degree refers to the polynomial after exact cancellation, not an upper bound obtained by adding degrees at every raw circuit gate. Two local identities expose the relevant cancellations. With $Z = X + \gamma H$ and $D_0 = ac + Z$,

$$A_0 = q^6\big[r^2 + \ell(2r - k) - 1\big] + 2q^3(\ell + r)Y + Y^2, \tag{22}$$

$$D_0^2 - \Delta c^2 = 2acZ + Z^2 - Hc^2, \tag{23}$$

$$A_1 = q^2\big[2acZ + Z^2 - Hc^2 - 1\big] + 2qD_0XH + (XH)^2. \tag{24}$$

These are exact integer polynomial identities. The cancellation-expanded certificate circuits implement them literally.

The degrees of $a, H, c, \Delta, I, U, \widehat{R}, V$ are respectively $2, 2, 2, 4, 5, 6, 3, 5$. Equations (22)–(24) then give the exact residual degrees

$$(\deg A_0, \deg A_1, \deg A_k, \deg A_t, \deg A_a, \deg A_s) = (11, 9, 4, 3, 12, 10).$$

Only A_a^2 contributes to total degree 24. Its leading homogeneous form is

$$\mathrm{in}_{24}(P_{15,24}) = 4X^4Y^6f^8u^2\big[(\eta + \zeta)T + hX\big]^2. \tag{25}$$

The coefficient of $X^6Y^6f^8u^2h^2$ is 4, independently of the fixed tuple. This proves *exact* degree 24, not merely degree at most 24.

The supplied evaluator uses 90 binary operations. An addition, subtraction, or multiplication is one operation, including multiplication by a small fixed numeral. Integer constants and references to earlier gates are free leaves. Every power is implemented by multiplication. Common subexpressions can be shared, and dead gates are pruned. This is the same source-level metric used for the comparisons; it does not count bit complexity or charge for the large coefficient values. The 79-gate shared residual evaluator gains six squarings and five additions for the single sum of squares.

The cancellation-expanded circuit uses more gates and is only a degree certificate. It is not substituted for the 90-operation evaluator when reporting cost. Both circuits compute the identical polynomial, as follows from the local identities and the literal downstream wiring.

8 Two fifteen-variable degree–cost tradeoffs

8.1 Degree 26 with 88 operations

Keep the fifteen variables and all common expressions of Section 3, but use

$$V = f^2(cT - R) - c, \qquad N_k = k - hE - R, \qquad A_{kt} = N_k W - q.$$

Define

$$P_{15,26} = A_0^2 + A_1^2 + A_{kt}^2 + A_a^2 + A_s^2. \tag{26}$$

The first two residuals again recover $Y = sq^3$ and $X = wq$. Since $W = qN_t$, the coupled residual is $q(N_kN_t - 1)$ and gives $N_kN_t = 1$. The auxiliary reconstruction is unchanged. These source factors satisfy the ordinary product equation (8), whose signed-unit proof supplies soundness. Canonical completeness gives $N_k = N_t = 1$ directly.

The residual degrees are $(11, 9, 7, 13, 10)$. With $q_0 = 3\beta j$, the degree-26 leading form is

$$4X^4Y^4f^8u^2q_0^6(q_0 - F)^2.$$

It is nonzero for the actual positive β. The literal evaluator has 88 gates. Keeping the degree-four R inside V costs two degrees in the final polynomial, while coupling the index and transport saves two gates. This makes the role of the last index reduction transparent.

8.2 Degree 30 with 79 operations

Use positive unknowns

$$J, F, \alpha, z, f, h, i, T, w, Y, r, \eta, \zeta, u, \gamma.$$

Here J is unrestricted within $\mathbb{Z}_{>0}$, $q = \beta J + 1$, and $X = wq$ is again a polynomial abbreviation. Keep Y directly supplied and use $V = f^2(cT - R) - c$. Put

$$N_1 = D_0^2 - \Delta c^2, \quad N_k = k - hE - R, \quad N_t = (K_0 + w)C + (q - F) - z(q - 1).$$

Then

$$P_{15,30} = A_0^2 + (N_1N_kN_t - 1)^2 + A_a^2 + A_s^2. \tag{27}$$

The first root alone forces $q^3 \mid Y$. The paired residual makes N_1, N_k, N_t integer units. Since $\Delta = (a + 2)^2 - 1$ is 0 or 3 modulo 4, $N_1 = -1$ is impossible modulo 4. Thus $N_1 = 1$ and $N_kN_t = 1$, giving the same source product. No coprimality restriction or main-root offset is needed. Completeness uses r and u as before and keeps $\gamma = \gamma_{\text{old}}$.

The four residual degrees are $(12, 15, 15, 10)$; the evaluator uses 79 gates. For an explicit exact-degree certificate, put

$$\begin{aligned} q_0 &= \beta J, & X_0 &= wq_0, & R_0 &= q_0^3(q_0 - F), \\ L_1 &= 8\gamma X_0^2 Y^3 k, & L_k &= -hX_0Y - R_0, & L_t &= w(q_0 - F - \alpha) - zq_0. \end{aligned}$$

The leading homogeneous form is

$$\mathrm{in}_{30}(P_{15,30}) = (L_1L_kL_t)^2 + (2X_0^2Y^2f^4uR_0)^2.$$

It is nonzero. In particular the monomial $w^4J^{10}Y^4f^8u^2F^2$ has coefficient $4\beta^{10}$, because the first square contains no u. This point offers most of the degree gain with four additional gates relative to the prior 44/75 equation, or three relative to the directly related 44/76 root-gap equation.

9 A twelve-variable companion of degree 629

The same integrality mechanism also improves the degree of a construction with fewer supplied variables. The result in this section uses twelve strictly positive integers and has exact degree 629, with 114 live binary arithmetic operations. Its three-equation version has maximum residual degree 622 and 109 operations, or 108 after the established positive transport shift. The comparison is with package 52's twelve-positive lean equation, of degree 897 and 108 operations. Thus the new point reduces degree by approximately 29.9%, at a cost of six operations. It does not reduce the number of supplied variables, and no global optimality or literature-wide novelty is claimed.

All notation in this section is local. The four fixed ports are the same complete authentic tuple (β, K_0, M_C, M_F) used elsewhere in this report; M_F is already shifted. In particular $\beta = 2^{590069} - 1$. Canonical completeness uses odd native exponent N, so its old coordinate satisfies $J_{\mathrm{old}} \equiv 1 \pmod 3$. We may therefore put $J_{\mathrm{old}} = 3J + 1$, retaining completeness with positive J by taking $N > 1$. This changes the parameterization, not the coefficient tuple. It gives $q \equiv 2 \pmod 3$. The point of this residue restriction is that the later divisor q is coprime to 3.

9.1 The fixed polynomial

The twelve supplied coordinates are

$$J, F, \alpha, z, h, i, T, s, w, \eta, \zeta, t \in \mathbb{Z}_{>0}.$$

Define the following polynomial abbreviations:

$$\begin{aligned}
j &= 3J + 1, & q &= \beta j + 1, & C &= q - F - \alpha,\\
X &= wq, & Y &= sq, & E &= XY,\\
k &= \eta + \zeta, & c &= kY + \eta, & a &= E + Y,\\
H &= 4a + 3, & \Delta &= a^2 + H = (a+2)^2 - 1, & &\\
\ell &= EkY, & A &= 1 + \ell(\ell + k), & G &= 1 + \Delta c^2,\\
b_0 &= X + ac, & & & &
\end{aligned}$$

$$R = \big((q-1)(q-F) + \alpha\big)(q^2 - 1) + (M_C + qM_F)j.$$

$$\begin{aligned}
N_k &= k - hE - R,\\
N_t &= (K_0 + w)C + q - F - z(q-1).
\end{aligned}$$

$$\begin{aligned}
I &= ic^2, & D &= 1 + \Delta I^2, & Q &= (\Delta I)^2,\\
d_0 &= Q - 1, & v &= cT, & b &= c + RD,\\
e &= Q(Dv^2 + b^2) - 1, & B_a &= 2Qvb. & &
\end{aligned}$$

The new scaled carrier is

$$\begin{aligned}
S &= q^2 H, & L &= q^2(1 + HIT),\\
W &= q^2 b_0 + H\big[q^2(t + b + \ell) + s\big]. & &
\end{aligned}$$

Let

$$\mathcal{N} = \prod_{\substack{g = \pm\sqrt{G},\ \tau = \pm\sqrt{A}\\ f = \pm\sqrt{D}}} \Big[d_0(W - Lg - S\tau)^2 - S^2 e + S^2 B_a f \Big]. \tag{28}$$

Every formal sign is included. Consequently this is an integer polynomial; no radicals or intermediate registers are additional unknowns. Formal signs are not being identified with independent field automorphisms after specialization. The equation is

$$\boxed{P_{12} = N_k N_t(1+\mathcal{N}) - 1 = 0.} \tag{29}$$

For an explicit division-free evaluator, put

$$\begin{aligned}
d_A &= d_0 S^2 A, & U_Q &= S^2 Q,\\
P_H &= U_Q D v^2, & U_H &= U_Q b^2,\\
e_H &= P_H + U_H - S^2, & j_H &= 4P_H U_H,\\
\mathcal{T} &= 2d_0(W^2 + L^2 G), & \kappa &= [d_0(W^2 - L^2 G)]^2,\\
\mathcal{A} &= 2(d_A + e_H), & \mathcal{B} &= (d_A - e_H)^2 - j_H,\\
a_r &= \mathcal{T} - \mathcal{A}, & b_r &= \mathcal{B} - \kappa,\\
v_r &= 16 d_A j_H - a_r(a_r\mathcal{T} + 2b_r), & u_r &= b_r^2 - a_r^2\kappa,\\
\mathcal{N} &= \kappa v_r^2 - u_r(\mathcal{T} v_r - u_r).
\end{aligned}$$

This is the inherited quadratic-remainder identity for the formal norm. The supplied JSON circuit gives every live gate and includes all multiplications by fixed numerals. The system version compares the final norm terms directly and imposes $N_k = N_t = 1$.

9.2 Bounds before recovering divisibility

The predecessor norm identity gives $\mathcal{N} \equiv (u_r - x_0 v_r)^2 \pmod 4$, where $x_0 = d_0(W^2 + L^2 G)$. At a root of (29), all three factors are integer units. The option $\mathcal{N} = -2$ is impossible modulo 4. Thus $\mathcal{N} = 0$ and $N_k = N_t = \varepsilon \in \{1, -1\}$.

The authentic native-mask inequalities applied to the transport unit give

$$C \geq 0, \qquad q - F = C + \alpha > 0, \qquad q \geq 16, \qquad 3q < R < q^4 - 2.$$

The boundary $C = 0$ is not discarded at this stage. We cannot yet use $q^3 \mid Y$ or $E > R$: the present definition is only $Y = sq$. Instead,

$$X, Y \geq q, \quad E \geq q^2, \quad k = hE + R + \varepsilon > R, \quad c > RY, \quad a + 2 < 2E, \quad I \geq c^2. \tag{30}$$

In bounds below, g, f, τ denote positive real square roots. Then $g < 2Ec$, $f < 2EI$, and the inherited elementary expansion gives

$$\Delta c^2 - A > k^2\big[EY(2Y^2 - 1) + Y^4\big] - 1 > 0.$$

In particular

$$1 < A < \Delta c^2 \leq \Delta I = \sqrt{Q}. \tag{31}$$

The earlier lean construction used $E > R$ in its outer-root estimate. Here it can be replaced by a consequence of $c > RY$. Since $D < 4E^2 I^2$, $c < I^2$, and $d_0 > \frac{1}{2}E^4 I^2$, we have

$$b + cf < (4R + 1)E^2 I^2, \qquad \frac{b + cf}{d_0} < \frac{9R}{E^2}.$$

On the other hand, the exact identity $(Ig)^2 - (cf)^2 = I^2 - c^2$ implies

$$Ig - cf = \frac{I^2 - c^2}{Ig + cf} > \frac{I}{8Ec} = \frac{ic}{8E}.$$

Finally $icE > RYE = RXY^2 > 72R$ by (30). Therefore

$$\frac{b + cf}{d_0} < \frac{9R}{E^2} < \frac{ic}{8E} < Ig - cf. \tag{32}$$

This is the extra estimate that makes divisibility recovery possible without assuming the scale one is trying to recover.

9.3 Integral roots force the missing scale

For a coherent auxiliary root y with $(f')^2 = D$, the norm relation is

$$d_0 y^2 = Q(vf' - b)^2 - 1.$$

The standard square estimate, also valid for complex y, gives

$$|y| < b + Tcf + T\frac{b + cf}{d_0} < b + ITg.$$

Consequently any root $x = \tau' + y$ of the inner polynomial has $|x| < \tau + b + ITg$. But the negative-main-root branch would give

$$\frac{W + Lg}{S} = b + ITg + \left(t + \frac{s}{q^2}\right) + \ell + \frac{b_0 + g}{H}.$$

This exceeds that upper bound: $t + s/q^2 > 0$, $\tau < \ell + k$, and $(b_0 + g)/H > c/3 > k$, since $Y \geq 16$. Actual quadratic conjugation would send an irrational positive-main-root branch to this forbidden branch. Hence g is a positive integer and $x = (W - Lg)/S$ is rational.

For completeness, the inherited inner-root reconstruction can be stated explicitly. Define

$$C_x = d_0(x^2 + A) - e,$$
$$F(x) = \left[C_x^2 + DB_a^2 - 4d_0^2 x^2 A\right]^2 - 4C_x^2 DB_a^2.$$

The norm implies $F(x) = 0$. If $C_x = 0$, eliminating x^2 would give

$$d_0^2 A^2 - d_0 eA + Q^2 v^2 b^2 D = 0,$$

which forces $Q \mid A(A-1)$, contrary to (31). Likewise $F(0) = 0$ would force $Q \mid (A-1)^2$, also impossible. Thus the denominators in

$$U_x = C_x^2 + DB_a^2 - 4d_0^2 x^2 A,$$
$$f' = -\frac{U_x}{2C_x B_a}, \qquad \tau' = \frac{C_x + B_a f'}{2d_0 x}, \qquad y = x - \tau'$$

are nonzero. The identity $F(x) = 0$ gives $(f')^2 = D$, $(\tau')^2 = A$, and the auxiliary Pell equation above. The rational square roots f', τ' are integers. The inherited rational Pell saturation lemma says that, for integers $m \geq 2, V$ and rational y,

$$m^2 V^2 - (m^2 - 1)y^2 = 1 \quad \Longrightarrow \quad y \in \mathbb{Z}.$$

Applying it with $m = \Delta I$ and $V = vf' - b$ shows that y, and hence x, is integral. The lemma's ordinary proof is preserved in the upstream proof files; it is not a newly supplied proof-assistant certificate.

Reduce $W = Lg + Sx$ modulo q^2. This gives $q^2 \mid Hs$. But $Y = sq$ gives $H \equiv 3 \pmod q$, while $q \equiv 2 \pmod 3$. Thus $\gcd(q, H) = 1$ and

$$q^2 \mid s, \qquad s_{\text{old}} = s/q^2 \in \mathbb{Z}_{>0}, \qquad Y = s_{\text{old}} q^3. \tag{33}$$

There is now a short soundness proof using the old equation directly. Let $\mathcal{N}_{\text{lean}}$ be package 52's unscaled twelve-variable lean norm, on the same fixed tuple. In the localization at q there is the exact identity

$$\mathcal{N} = q^{32} \mathcal{N}_{\text{lean}}\left(t + \frac{s}{q^2}\right), \qquad Y = (s/q^2)q^3. \tag{34}$$

Each of eight factors gains q^4. Its carrier degree is at most 16, so all denominators clear. By (33), the mapped old coordinates s_{old} and $t_{\text{old}} = t + s_{\text{old}}$ are positive integers. The index and transport factors are unchanged. Thus a new root produces a root of the established old positive equation. Its compiler and K215 calibration recover a finite ZFC refutation.

9.4 Completeness and coefficient-preserving equivalence

Choose canonical normalized compiler witnesses with odd $N > 1$. Keep the retained coordinates, put $J = (J_{\mathrm{old}} - 1)/3$, $s = s_{\mathrm{old}} q^2$, and set

$$t = \gamma + ITg + \tau + y - b - \ell - s_{\mathrm{old}}.$$

All coordinates are integral. The auxiliary Pell equation gives $y \geq b - vf$, and $Ig > cf$. Also

$$\tau - \ell = \frac{\ell k + 1}{\tau + \ell} > \frac{k}{3}, \qquad \frac{k}{3} > \frac{XY}{3} > \frac{Y}{q^3} = s_{\mathrm{old}}.$$

Here $\ell > k$, $\tau < \ell + k$, $k > XY$, and $X \geq q \geq 16$ are canonical compiler inequalities. Therefore

$$t \geq \gamma + T(Ig - cf) + (\tau - \ell) - s_{\mathrm{old}} > 0.$$

A coherent factor in (28) vanishes, and $N_k = N_t = 1$. This proves both existence implications. Composing with the inherited syntactic-refutation calibration yields, by ordinary mathematics formalizable in ZFC,

$$\mathrm{ZFC} \vdash [P_{12} \text{ has no positive integer zero}] \longleftrightarrow \mathrm{Con(ZFC)}.$$

No truth value of either equivalent assertion is proved. The separately supplied system replaces the product by $N_k = N_t = 1$ and $\mathcal{N} = 0$. Its shifted transport version uses $(K_0 + w)C = F + z(q - 1)$; the established positive transport shift preserves existence after the scale has been recovered.

9.5 Exact degree and reproducible evidence

Set the leading homogeneous forms

$$\begin{aligned}
q_0 &= 3\beta J, & X_0 &= wq_0, & Y_0 &= sq_0, & E_0 &= X_0 Y_0,\\
k_0 &= \eta + \zeta, & c_0 &= k_0 Y_0, & I_0 &= ic_0^2, & \ell_0 &= E_0 c_0,\\
H_0 &= 4E_0, & Q_0 &= E_0^4 I_0^2, & R_0 &= q_0^3(q_0 - F), & &\\
B_0 &= R_0 E_0^2 I_0^2, & V_0 &= I_0 T \ell_0. & & & &
\end{aligned}$$

The respective degrees of $H_0, \ell_0, Q_0, B_0, V_0, c_0$ are $4, 7, 30, 26, 15, 3$. The exact leading cancellations in the unscaled lean norm give the following eight factor degrees:

$$79, 79, 79, 79, 71, 71, 67, 65.$$

Their sum is 590. Four opposite f/g branches have leading form $-\operatorname{sgn}(g) 4H_0^2 Q_0 B_0 V_0$; the two matched branches with negative τ have leading form $4H_0^2 Q_0 B_0 \ell_0$; the other two matched branches have leading forms

$$H_0^2 Q_0 B_0 c_0, \qquad H_0^2 Q_0 B_0 \left(2t - k_0 - \frac{iTc_0}{E_0}\right).$$

Their product is

$$2^{12} H_0^{16} Q_0^8 B_0^8 V_0^4 \ell_0^2 c_0 \left(2t - k_0 - \frac{iTc_0}{E_0}\right). \tag{35}$$

The displayed denominator cancels against preceding factors. The result is a nonzero polynomial, since t is independent. These are the same exact lean-norm cancellations as in the bundled predecessor proof, now with $\deg X = \deg Y = 2$; in particular the cancellation uses $(Ig)^2 - (cf)^2 = I^2 - c^2$, not independent treatment of these square roots.

The shift s/q^2 in (34) has degree -1, below $\deg t = 1$, so it does not change (35). The factor q^{32} adds 32 degrees, giving $\deg \mathcal{N} = 622$. The index factor has nonzero degree-5 leader $-hE_0$, and the transport factor has nonzero degree-2 leader $w(q_0 - F - \alpha) - zq_0$. Hence

$$\deg P_{12} = 622 + 5 + 2 = 629.$$

This is uniform for every authentic positive β satisfying the stated coefficient and completeness interface.

The companion checker performs 60 modular formal-norm multiplications, 59 valid modular checks of (34), and six complete univariate polynomial computations: three with formal-port fixtures and three with the actual fixed tuple. All six give degrees 622 and 629. It also checks all four compressed and uncompressed coefficient hashes and all fourteen bundled source-file pins. These are finite algebra diagnostics; the unbounded equivalence rests on the preceding argument and the inherited compiler theorems. The files `twelve/upstream/LEAN_PROOF.md`, `INNER_RECOVERY_PROOF.md`, `ELEVEN_POSITIVE_PROOF.md`, and `ORDINARY_BLANK_PROOF.md` preserve those ordinary mathematical boundaries explicitly.

10 The fixed coefficients and their provenance

All four new equations use the complete compact native tuple from source 44 [1]. The binding is

$$\beta = B - 1, \quad K_0 = D_C + BD_R, \quad M_C = M_C, \quad M_F = M_{F,\mathrm{native}} + B - 1, \qquad B = 2^{590069}.$$

The last shift is part of the source interface. Replacing it by the native high mask changes the polynomial and invalidates the stated theorem. Nor can individual ports be selected from different historical packages.

For completeness, the following is a finite specification of the numerals. Let $\rho = 2^{67}$, $L = 8807$, so $B = \rho^L$. The ordered list of 943 literal Wang tiles is supplied in the archive. Its edge labels (w, n, e, s) have respective widths $8, 9, 8, 9$. Set

$$v_t = w + 2^8 n + 2^{17} e + 2^{25} s + 127\, 2^{34}, \qquad c_t = 1 + 2^{10}(2^{41} - v_t).$$

Let $\mathcal{E}$ consist of selector positions $0, \dots, 942$, west copies $1023, \dots, 1031$, north copies $1013, \dots, 1022$, east copies $1003, \dots, 1011$, south copies $983, \dots, 992$, dummy positions $1063, 1064$, and anchors $1095, 2195, 3296$. For selectors use the coefficients c_t. The west payload bits are $0, \dots, 7, 34$; north bits $8, \dots, 16, 35$; east bits $17, \dots, 24, 36$; south bits $25, \dots, 33, 37$; the anchors are bits $38, 39, 40$. A copy or anchor carrying bit j receives coefficient 2^{10+j}. Both dummy coefficients are zero. Define

$$\begin{aligned}
\mu &= 2^{51} - 2, \\
D_C &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \sum_{e \in \mathcal{E}} c_e(\rho^{4393-e} + \rho^{5495-e}), \\
D_R &= \rho^{42}, \\
Z_{\mathrm{pad}} &= (2^{65} - 1) \sum_{e=8792}^{8804} \rho^e + (2^{19} - 1)\rho^{8805}, \\
M_C &= B - 1 - \sum_{\substack{e \in \mathcal{E} \\ e \neq 1}} \rho^e - 2\rho^{1063}, \\
M_{F,\mathrm{native}} &= 4 + \mu(\rho^{4393} + \rho^{5495}) + \sum_{j=0}^{9} \rho^{1013+j} + \sum_{j=0}^{8} \rho^{1065+j} + \rho^{2195} + \rho^{3296} + Z_{\mathrm{pad}}.
\end{aligned}$$

All summation limits and exponents are fixed numerals. The archive supplies both the literal tile list and the literal 430-action K215 transition table from which the coefficient builder reconstructs

Port	Bit length	Raw bytes	Gzip bytes
β	590069	73759	108
K_0	592884	74111	8940
M_C	590069	73759	193
M_F (shifted)	590070	73759	204
Total		295388	9445

Table 2: Materialized fixed ports. These are numeral-port sizes, not a claim about the coefficients of a fully expanded polynomial.

it, without running the machine. Thus the specification makes no reference to whether the machine eventually halts.

Each gzip payload contains an unsigned little-endian integer. The binding records both compressed and raw SHA-256 digests. The principal circuit aliases are `K0` $\mapsto$ `K` and `MF` $\mapsto$ `MF_source`. The other two aliases are unchanged. The twelve-variable circuits use the same map.

The copied adapter proves the complement-clause lemma, native alignment, carry and population bounds, and the actual-rank padding controller. Its parameters are $d = 67 \cdot 8807 = 590069$ and $\lambda = 581197$. In particular, the controller supplies powers of an odd prime for both padded dimensions, so the odd-N argument used in the new proof is part of this exact tuple's completeness theorem. The native low mask has residue 2 modulo 4 and the canonical low-bit disjointness gives $C \equiv 1 \pmod 4$, whence $R \equiv 3 \pmod 4$. These facts are not presumed for arbitrary integer ports.

The independent inherited coefficient auditor was rerun. It authenticated the source files and checked 66,062 native basis identities, 38,663 one-bit mutation diagnostics, and 87,380 exhaustive small complement models, as well as the concrete bounds and controller residues. Finite audits verify that the implemented constants satisfy the finite parts of the compiler interface; the unbounded semantic theorem remains the preserved ordinary mathematical proof.

11 Domains and what "no solutions" means

The compact result is stated over strictly positive integers. This is a standard Diophantine convention, but the domain must be written explicitly. If $\mathbb{N}$ denotes $\{0, 1, 2, \ldots\}$, the polynomial

$$P_{\mathbb{N}}(x_1, \ldots, x_{15}) = P_{15,24}(1 + x_1, \ldots, 1 + x_{15})$$

has the same existence statement over $\mathbb{N}^{15}$. Its exact degree is still 24, since translation preserves the top homogeneous component.

For a completely unrestricted integer equation, let

$$Q_j = 1 + z_{j1}^2 + z_{j2}^2 + z_{j3}^2 + z_{j4}^2, \qquad 1 \le j \le 15,$$

and put $P_{\mathbb{Z}} = P_{15,24}(Q_1, \ldots, Q_{15})$. Lagrange's four-square theorem makes every positive integer attainable as a Q_j. Therefore

$$\neg\exists \mathbf{z} \in \mathbb{Z}^{60}\ P_{\mathbb{Z}}(\mathbf{z}) = 0 \quad \longleftrightarrow \quad \mathrm{Con(ZFC)}$$

is again provable in ZFC. The polynomial has exact degree 48: the degree is at most twice 24, and setting three coordinates in every block to zero leaves an injective substitution $x_j \mapsto z_{j1}^2$ in the nonzero top form. This translation is supplied for domain clarity, not claimed as a competitive unrestricted-integer frontier. In particular, this work does not improve the supplied ten-integer-variable constructions.

12 Reproducibility and limits of verification

The archive is designed to be usable independently of the working checkout. The main deliverables are this PDF and its LaTeX source, the literal JSON circuits, Python builders and checkers, the complete coefficient payloads, source provenance, and the inherited proof files. Python verification uses only the standard library. The scripts locate files relative to themselves and also work under `python3 -O`; correctness checks use explicit exceptions rather than removable `assert` statements.

The main checks have different logical roles:

1. **Exact algebra and degree.** Five local sparse polynomial identities prove the cancellation and index-reduction rewrites. Leading homogeneous forms are propagated exactly through the certificate circuits. A completely vanishing leading form causes failure, rather than silently accepting an uncertain degree. The resulting forms include (25) and the two tradeoff leaders.

2. **Literal circuit integrity.** All references are defined before use; every charged operator is binary $+, -$, or $\times$; every advertised unknown and port is live; all operation counts are computed from retained gates. Builders regenerate the published circuits deterministically.

3. **Independent transcription.** A separate checker imports no project builder or algebra implementation. It authenticates the actual coefficient payloads, compares independently transcribed formulas with the primary and certificate circuits in 7,344 scalar residual/output checks, and performs 72 full univariate finite-field polynomial comparisons. These rays also provide matching degree lower bounds at the actual ports.

4. **Twelve-variable companion.** Its independent checker performs 60 formal-norm multiplication checks, 59 valid old-norm substitution checks, and six full polynomial-ray calculations, including three at the authentic tuple. It authenticates its fourteen inherited source pins.

5. **Coefficient and source authentication.** The main algebra verifier authenticates 51 copied source files. The independent coefficient auditor reconstructs the finite native constraints and controller checks. Hashes detect changed files; they do not prove the semantic content of a proof document.

The independent mathematical reviews were produced by separate AI-assisted review agents. They checked the new soundness and completeness reductions, the positive margins, the coefficient interface, the twelve-variable pre-divisibility bounds, and the degree calculations. These are internal reviews, not external peer review or proof-assistant certification. Their notes are retained so that a human reviewer can examine both the arguments and their stated boundaries.

No test searches for a refutation of ZFC, assumes that a zero exists, or infers consistency from a failure to find one. Evaluating randomly chosen tuples does not establish an unbounded existence equivalence. That equivalence rests on the displayed reductions and the inherited compiler theorems. A full formalization would need to encode and check those reductions against the exact bound constants and calibrated syntax.

13 What is new, and what remains open

The core new step is the reuse of monic norm roots to carry divisibility conditions. Directly supplying Y removes the repeated degree cost of sq^3. A second carrier directly supplies X after the first has made its offset coprime to the modulus. A separate index equation then lowers the auxiliary expression's degree. The squared-modulus quotient, the ordinary strong-rank theorem, the root-gap technique itself, the compact coefficient tuple, and the twelve-variable formal-norm evaluator are credited inherited ingredients. The proof is an existence reduction

and may choose fresh auxiliary witnesses; it is not a same-coordinate identity with the entire predecessor polynomial.

The twelve-variable result shows that the idea survives elimination of explicit square-root witnesses, but it requires a new bound before the missing scale is recovered. That order matters: using $E > R$ at that stage would have assumed what the construction is supposed to prove. The bound in Section 9 instead uses $c > RY$ and $E \geq q^2$.

Several concrete directions remain. One can investigate whether another existing monic relation can carry the remaining quotient conditions, apply the lower-degree parent to selective witness elimination, and formalize the two scale-recovery lemmas and the canonical positive-gap construction. The literal circuits also leave room for an operation-count optimization that does not change the mathematical polynomial. None of these further improvements is included in the stated result.

There is no proof here that degree 24 is minimal with fifteen positive unknowns, or that these equations are simplest under an unspecified global metric. What is established is a concrete new set of degree–cost points, with fixed numerals, explicit equations, two-way reductions, and reproducible arithmetic evidence, compared with a precisely identified source snapshot.

A Reproduction commands and archive map

From the unpacked archive root, the single command `python3 verify_all.py` runs the complete finite audit. The individual commands are:

```
python3 code/build.py
python3 code/domain_translations.py
python3 code/verify.py
python3 code/independent_check.py
python3 twelve/build_p12.py
python3 twelve/check_p12.py
python3 coefficients/audit_coefficients.py
```

The top-level runner executes these checks in order and records their results. The LaTeX document can be rebuilt with:

```
cd report
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The second pass resolves cross-references and the contents. The twelve-variable section is included from its own source file. The PDF was rendered and visually inspected before packaging.

Directory or file	Purpose
`report/`	Complete new paper, LaTeX source and rendered PDF.
`circuits/`	Three primary evaluators, their degree certificates, and explicit domain translations.
`code/`	Deterministic builders and exact/independent checkers.
`certificates/`	Machine-readable receipts and homogeneous forms.
`coefficients/`	Complete compact tuple, source data, native proof, builder, inherited independent auditor, and notices.
`twelve/`	Companion equation, systems, proof, checker, LaTeX section, and fourteen pinned upstream sources.
`upstream/`	Inherited ordinary kernel, Pell lemmas, comparison proofs, and semantic status documents.
`review/`	Explicitly labeled internal AI-assisted review notes.
`PROVENANCE.json`	Repository snapshot, exact source paths and digests.
`MANIFEST.sha256`	Hash inventory of the delivered files.

References

[1] ProveIt source 44, *Complement-packed native clauses: a smaller complete coefficient tuple*. Pinned source at the repository snapshot identified in Section 1; full proof and payloads reproduced in `coefficients/`.
https://github.com/VladimirReshetnikov/ProveIt/tree/6d375f2d2deac0519e659bbce5b48087023c1430/Computability/DiophantineConsistency/con-zfc-diophantine-forms/44-b4-sparse/native

[2] ProveIt source 49, *Support-aware Pell circuits*, section on the fifteen-positive root-gap constructions. Preserved as `upstream/OFFSET44_PRIOR.tex`.
https://github.com/VladimirReshetnikov/ProveIt/tree/6d375f2d2deac0519e659bbce5b48087023c1430/Computability/DiophantineConsistency/con-zfc-diophantine-forms/49-support-pell

[3] ProveIt source 47, *Minimum-norm consistency polynomials*, supplement with the degree-44, 75-operation fifteen-positive equation. Proof and literal circuit preserved under `upstream/MINIMA44_PRIOR*`.
https://github.com/VladimirReshetnikov/ProveIt/tree/6d375f2d2deac0519e659bbce5b48087023c1430/Computability/DiophantineConsistency/con-zfc-diophantine-forms/47-sparse-support/minima

[4] ProveIt source 51, *A 66-operation consistency polynomial by a squared-modulus quotient*. Preserved as `upstream/SQUARED_MODULUS_PROOF.md`.
https://github.com/VladimirReshetnikov/ProveIt/tree/6d375f2d2deac0519e659bbce5b48087023c1430/Computability/DiophantineConsistency/con-zfc-diophantine-forms/51-sparse-arith

[5] ProveIt source 52, *Sparse Pell norms and low-variable consistency equations*. The twelve-positive lean proof, evaluator, and needed recovery lemmas are preserved in `twelve/upstream/`.
https://github.com/VladimirReshetnikov/ProveIt/tree/6d375f2d2deac0519e659bbce5b48087023c1430/Computability/DiophantineConsistency/con-zfc-diophantine-forms/52-sparse-pell

[6] ProveIt, *K215: 215 working states for ZFC consistency*, status at the pinned snapshot. Preserved as `upstream/K215_README.md`.
https://github.com/VladimirReshetnikov/ProveIt/tree/6d375f2d2deac0519e659bbce5b48087023c1430/Computability/BusyBeaver/Research/ZFC/candidate-k215

[7] Zhi-Wei Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, preprint, arXiv:2609.01594, version 2, 2026.
https://arxiv.org/html/2609.01594v2

[8] Jonas Bayer, Marco David, Malte Haßler, Yuri Matiyasevich, and Dierk Schleicher, *Diophantine equations over $\mathbb{Z}$: universal bounds and parallel formalization*, arXiv:2506.20909, version 3, 2026.
https://arxiv.org/html/2506.20909v3

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https://doi.org/10.4230/LIPIcs.ITP.2025.3